Generalizable Robotic Intelligence through Structured and Generative Learning

Research Statement for PhD Application

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1 Introduction

I have always been deeply fascinated by the fundamental technologies that underpin robotics and autonomous systems. Motivated by the limitations of how current robots operate, I aim to develop generalizable solutions that can adapt to various tasks and environments, enhancing the capabilities of robots in real-world scenarios. I believe that endowing robots with generalizable decision-making capabilities is key to unlocking their transformative impact on society. This challenge is not merely a matter of improving performance metrics but requires a deeper understanding of model utilization and systematic integration of expressive models into robotic systems. My research tackles this challenge by focusing on two complementary themes: leveraging structural insights to guide model and policy generalization, and incorporating expressive models, such as generative world models, to enhance adaptability across diverse robotic tasks.

2 Previous Research

Over the past two years, my research has focused on learning-based methods to compensate for residual physics and disturbances in robotic dynamics, particularly under aggressive motions. I have developed several frameworks from both planning and control perspectives. On the planning side, my work involves fine-tuning reference trajectories [1] to minimize control errors and leveraging neural ordinary differential equations (neural ODEs) to model and mitigate residual effects along optimized trajectories [2]. For control, I have explored online learning of disturbances using meta-learned representations that enable real-time adaptation to novel disturbances [3]. Additionally, I have investigated the decoupling of disturbances into state-dependent and time-varying components using Chebyshev polynomials [4]. Through these efforts, I observed that model generalizability stems from both **representation capacity** and **structured design**, with the latter benefiting significantly from principles in control theory. Building on this insight, I proposed a feedback-augmented neural ODE framework that learns latent dynamics while incorporating online state feedback to calibrate model rollouts, resulting in more generalizable online predictions [5]. Despite promising results, these models are not yet directly applicable to more complex tasks, such as locomotion of articulated robots, long-horizon planning, or decision-making in manipulation. This motivates my ongoing efforts further to enhance the generalizability and scalability of learning-based robotic systems.

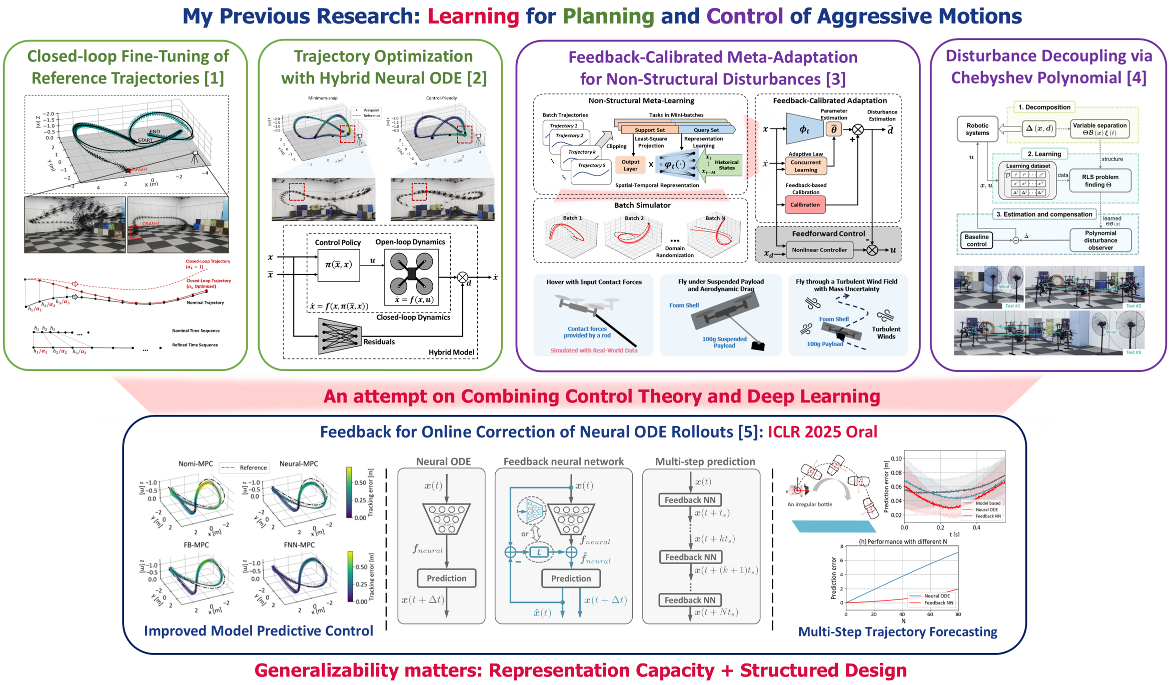

Figure 1: Overview of my previous research on learning-based planning and control, with extensions toward applying control theory to the design of deep learning models.

3 Current Work

Generative models, such as diffusion-based [6], [7] and flow-based architectures [8], have demonstrated remarkable compositional generalization, enabling the generation of high-quality samples from prior distributions. When conditioned on specific tasks or environments, these models exhibit strong adaptability and generalizability in robotic applications [9], [10], [11]. However, integrating generative models into low-level control frameworks remains challenging due to high inference latency and the need to enforce soft dynamic feasibility constraints. To address these challenges, I am working on approaches to embed generative models within model predictive control (MPC) frameworks, while ensuring that the overall computational complexity remains compatible with real-time MPC requirements.

4 Future Directions

My future research will focus on learning generalizable solutions for robotic applications. Optimal control (OC) and reinforcement learning (RL) are two promising approaches to solve control, planning, and higher-level decision-making problems in robotics, in which reinforcement learning enjoys higher flexibility for complex tasks while optimal control is more efficient and interpretable [12]. Both paradigms fundamentally rely on models to predict system dynamics and support planning. Additionally, RL introduces policies as central components for action generation. Therefore, enhancing model generalization is crucial across both OC and RL frameworks, while policy generalization is uniquely critical in RL settings. To address these challenges, I plan to pursue two interrelated directions: (i) designing structural abstractions that enable adaptable representations across diverse environments, tasks, and robot morphologies; and (ii) leveraging capable and expressive models, such as generative world models, to capture rich task and environment variations and support scalable decision-making.

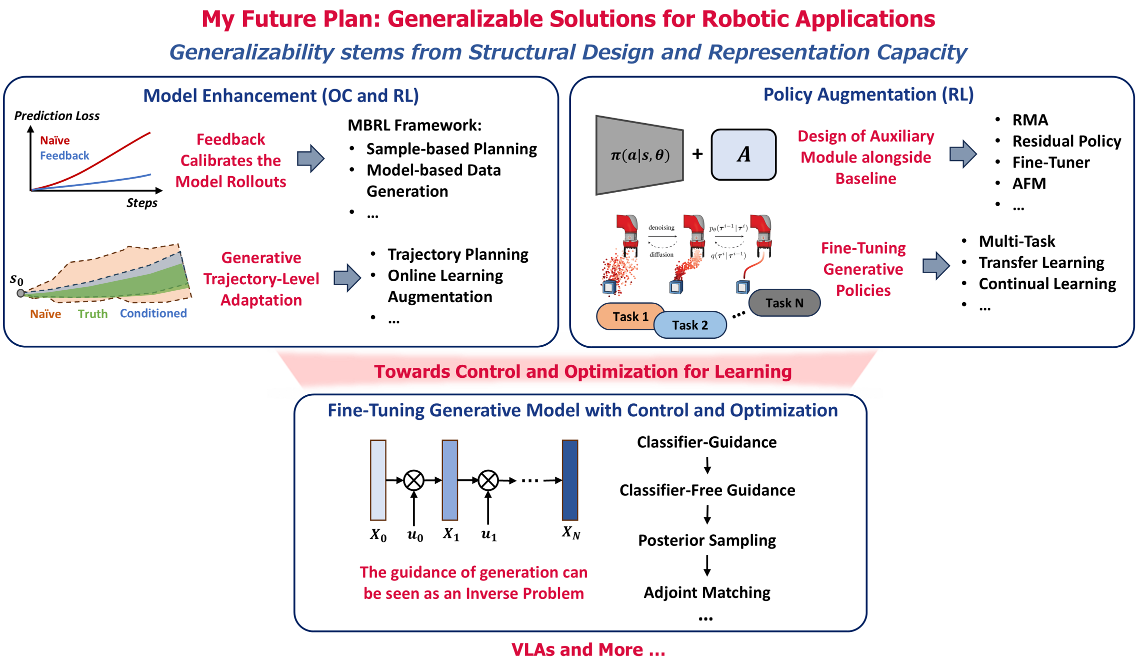

Figure 2: Research roadmap illustrating my future work on generalizable modeling in robotics, with extensions toward control-guided generative models and advanced embodied intelligence through vision-language-action frameworks.

4.1 Model Enhancement

Models are the core of robotic systems, as they provide the necessary representations for planning and control. As mentioned in Section 2, integrating feedback mechanisms from control theory can be a good structural abstraction to improve model generalizability [5]. For model-based RL frameworks [13], [14], it can effectively calibrate model rollouts during online inference, leading to more reliable sample-based planning and data generation. On the other hand, leveraging high-capacity models such as generative models further enhances predictive performance [9], [11], [15]. Combining these models with online learning techniques opens up promising directions for trajectory-level domain adaptation, thereby improving the generalizability of trajectory planning frameworks.

4.2 Policy Augmentation

Building strong policies that can adapt to diverse tasks and environments is a key challenge in robotics. Structural modification can be made to the baseline policy by integrating it with auxiliary module, such as an environment encoder [16], a residual policy [17], a fine-tuner [18], [19] or an adaptive mechanism [20], [21]. Alongside these structural modifications, high-capacity models such as generative models can be used to augment the policy [22]. Fine-tuning such policies with task-specific guidance can unlock the generalizability beyond learning from demonstration [23], leading to potentially multi-task, continual learning, and transfer learning capabilities.

4.3 Generative Model Guidance and Beyond

Orthogonal to the above two directions, I am also interested in fine-tuning generative models using control and optimization. The guidance of diffusion-like generative models can be seen as an inverse problem, where a goal condition or posterior distribution is desired to be achieved. Beyond classifier [24] and classifier-free guidance [25], recent works [26], [27] have shown that the optimal control framework can be used to guide the generative model. This opens up the new possibility of integrating generative models with control and optimization. In addition, exploring the integration of more advanced embodied foundation models, such as Vision-Language-Action Models [28], represents a promising direction. These models combine large-scale perception, reasoning, and decision-making capabilities, potentially enabling robots to perform a broader range of tasks with greater autonomy and adaptability in real-world environments.

5 Conclusion

In summary, I seek to enhance the adaptability and generalizability of robotic systems across diverse tasks and environments, leveraging structural insights and high-capacity models. Besides, I aim to explore how the ideas from the control society can be applied to the broader field of machine learning. I believe that these directions will not only advance the field of robotics but also contribute to the learning society. I am excited about the potential of my research to make a significant impact in the field of robotics and to contribute to the development of more capable and versatile robotic systems, as well as impressive machine learning models.

Overall, we are working towards a better future, which can be seen as an inverse problem in the sense that we are endeavoring to transform our world towards some optimal conditions. Let's try to solve such inverse problems that encourage human progress.

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